

Busy Snail Filament Changer (BSFC) BOM V.2026-02-04 for 4x1kg spool units changer, totals

	Item	Qty	Info	2026-02 sample link
	Heat set insert M3xD5xL4	68	Same as Voron BOM size	
	Flat Head Cap Screw M3x6	4		
	Flat Head Cap Screw M3x8	24		
	Socket Head Cap Screw M3x6	8		
	Socket Head Cap Screw M3x10	8		
	Socket Head Cap Screw M3x14	4		
	Socket Head Cap Screw M3x20	16		
	Socket Head Cap Screw M3x25	4		
	Socket Head Cap Screw M3x30	12		
	Self tapping screw M2x5	6		
	Self tapping screw M2x8 flat tip	36		<u>SELF TAPPING FLAT TIP</u>
	Nut M12	4	M12x1.75 DIN934	<u>NUT</u>
	Washer M12	4	M12x24x2.5 DIN125	<u>WASHER</u>
	Threaded rod M12	2	M12x1.75 DIN975, length depends on the changer and printer setups	<u>THREADED ROD</u>
	BMG 17 teeth extruder gear	4	Extruder drive gear with grub screw for 5mm stepper shaft	<u>BMG-17 GEAR</u>
	Ball bearing MR63zz	8+1	Included 8 in 4x HGX-lite extruder part kits, 1 extra needed	<u>HGX-LITE KIT / BEARING MR63zz</u>
	Thumb screw assy with spring	4+4	Included 4 in 4x HGX-lite extruder part kits, 4 extras needed	<u>HGX-LITE KIT / THUMB SCREW ASSY</u>
	Large extruder gear	8	Included in 4x HGX-lite extruder part kits	<u>HGX-LITE KIT</u>
	Needle bearing 3x5x6.5mm	8	Included in 4x HGX-lite extruder part kits	<u>HGX-LITE KIT</u>
	Bearing pin 3x16mm	8	Included in 4x HGX-lite extruder part kits	<u>HGX-LITE KIT</u>
	Bearing pin 3x12mm	12		<u>BEARING PIN</u>
	Tension spring D6xL20xWD0.4	1	Tension spring for buffer slide (Total length with eyelets 6+8+6=20mm)	<u>TENSION SPRING</u>
	PTFE tube OD4mm ID2mm	1m		<u>PTFE TUBE</u>
	PTFE tube OD6mm ID4mm	3m	3 meter continuous lot	<u>PTFE TUBE</u>
	Tube collet 4mm	5		<u>TUBE COLLET 4mm</u>
	Tube collet 6mm	9		<u>TUBE COLLET 6mm</u>
	Cable ties, small	20	1.8mm tie width, 3.5mm eye width	<u>SMALL CABLE TIE</u>
	Bicycle inner tube	≈0.3m	Optimal flat width ≈40-45mm to be stretched on a 32mm diameter roller, 8x35mm pieces	

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	Stepper motor NEMA17 L=38mm	4	Optimal L=38(40)mm can be longer but affects wiring length and units sideways stacking fitment	NEMA17
	Led SK6182 RGBW	4		LED
	Micro switch D2F	9		MICRO SWITCH
	Micro switch D2F-L	2		MICRO SWITCH
	JST PH2.0 3-m-pin housing	12	Switch and led inlet drive unit	JST PH2.0 HOUSING
	JST PH2.0 4-m-pin housing	5	Motor inlet drive unit (4) and hub/buffer (1)	JST PH2.0 HOUSING
	JST PH2.0 6-fm-pin housing	4	Stepper motor connector	JST PH2.0 HOUSING
	JST PH2.0 3-fm-pin housing	12	Wirings on drive unit ends	JST PH2.0 HOUSING
	JST PH2.0 4-fm-pin housing	5	Wirings on drive unit and hub/buffer ends	JST PH2.0 HOUSING
	JST PH2.0 male pin	56		JST PH2.0 MALE PIN
	JST PH2.0 female pin	72		JST PH2.0 FEMALE PIN
	FEP wire UL1332 26AWG	≈25m	OD max 1.1mm to fit wire bundles to channels in 3d printed parts at some tight places	UL1332 FEP WIRE
	Controller board*	1	For 4 spool units changer: a board with 4 stepper drivers, minimum 11 switch inputs and rgb	BTT MMB CAN V2

*Controller board box and wiring instructions supports Btt Mmb Can V1 and V2 boards. Wiring connectors on board end are not included in this BOM.